Recommender Systems II

Statistical Data Mining II

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Outline

- Introduction
- User based CF
- Item based CF

Introduction

Overall Goals of a recommender system:

1. Prediction version of the problem:
   - Predict the rating value for user-item combination.
   - Training data is available – user preference information.
   - Matrix Completion problem: specified values (observed) are used to predict the missing (unobserved).

2. Ranking version of the problem:
   - recommend the ”top k” items for a particular user

     or

   - recommend the “top k” users to target for a particular item.

Introduction: Operational and Technical Goals

1. **Relevance**: recommend relevant items to the user at hand. Although the primary goal, it is not sufficient in isolation.
2. **Novelty**: Ranking items that have not been seen by the user in the past. (popular movies by a liked genre, or repeated recommendations of popular items. – also reduces diversity)
3. **Serendipity:** The items recommended are somewhat unexpected, user surprised to learn of item.

   e.g., Indian restaurant lover – a recommendation for a new Indian restaurant is not surprising. On the other hand, Ethopian restaurant would be surprising.
4. **Increasing recommendation diversity**: “top-k” items should not be too similar – increases the chance that the user will not like any of these!

Introduction: Soft Goals

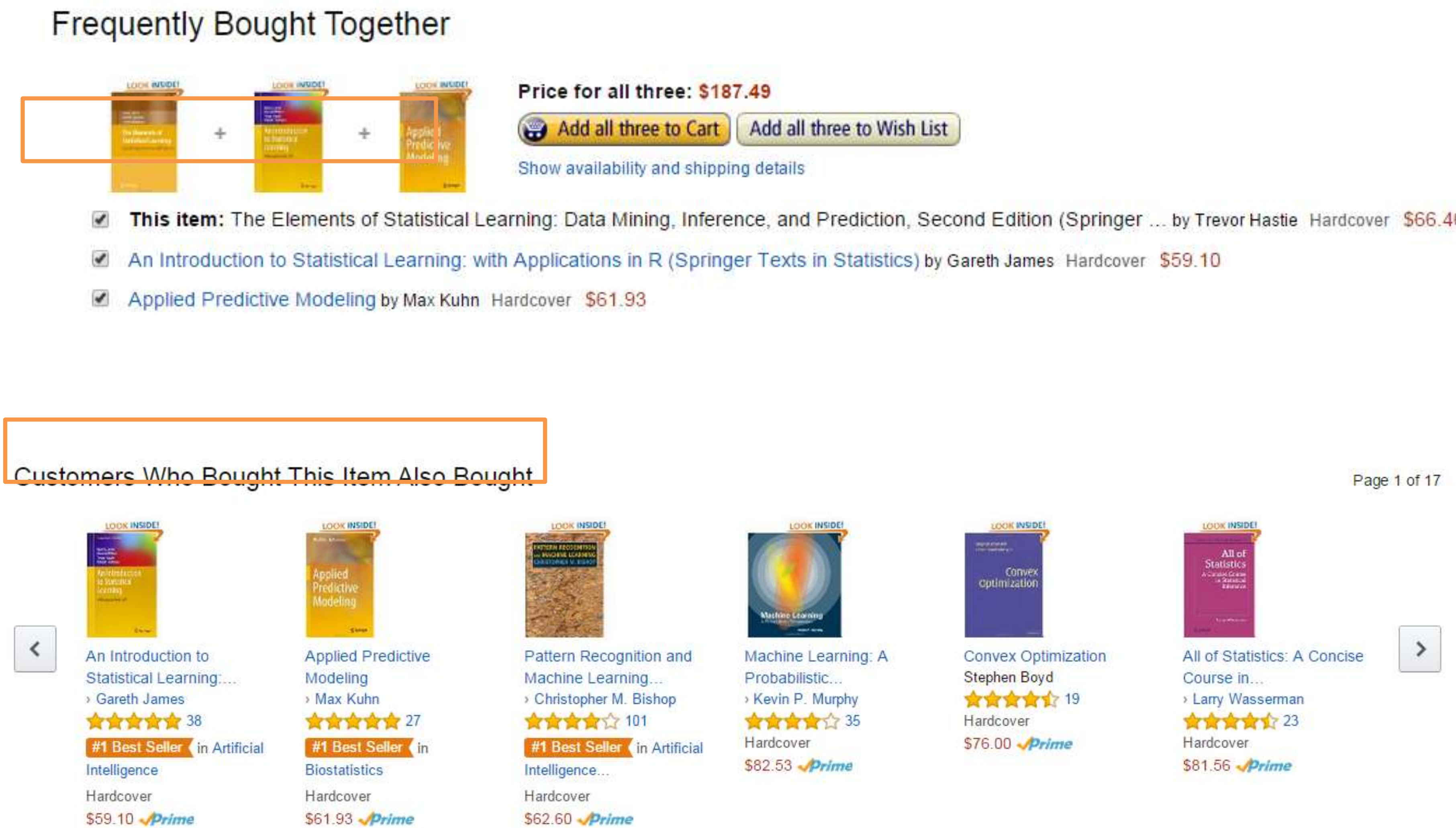
1. Repeat satisfaction with relevant recommendations. Improves loyalty and overall satisfaction.
2. Providing the user explanation for why a particular item is recommended is often useful.

Amazon: a combination of ratings, browsing behavior and buying behaviors. Often explanations are provided.

Netflix: provides a good job of giving explanations.

Introduction: Non-personalized

Introduction: Personalized



http://www.amazon.com/Elements-Statistical-Learning-Prediction-Statistics/dp/0387848576/ref=sr_1_1?s=books&ie=UTF8&qid=1416592593&sr=1-1&keywords=elements+of+statistical+learning

Introduction

- Netflix: 2/3 of the movies watched are recommended
- Google News: recommendations generate 38% more clickthrough
- Amazon: 35% sales from recommendations

*new wave of competitions…..
Netflix – 1 million dollar prize to beat algorithm by 10%.

Introduction: Real Examples

	GLADITOR	GODFTHER	BEN-HUR	GOODFELLAS	SCARFACE	SPARTACUS
U_1	1			5		2
U_2		5			4	
U_3	5	3		1		
U_4			3			4
U_5				3	5	
U_6	5		4			

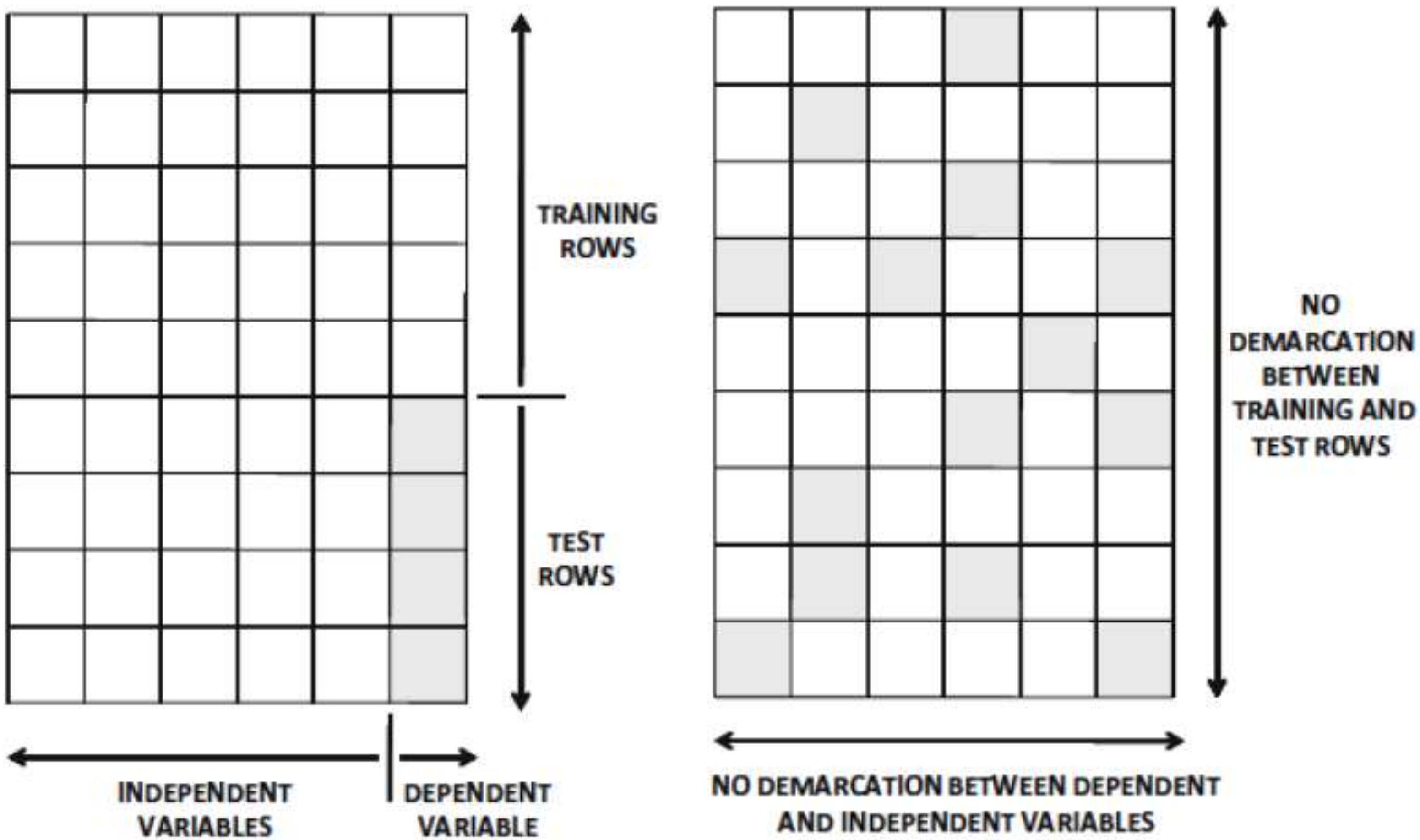
(a) Ordered ratings

	GLADIATOR	GODFATHER	BEN-HUR	GOODFELLAS	SCARFACE	SPARTACUS
U_1	1			1		1
U_2		1			1	
U_3	1	1		1		
U_4			1			1
U_5				1	1	
U_6	1		1			

(b) Unary ratings

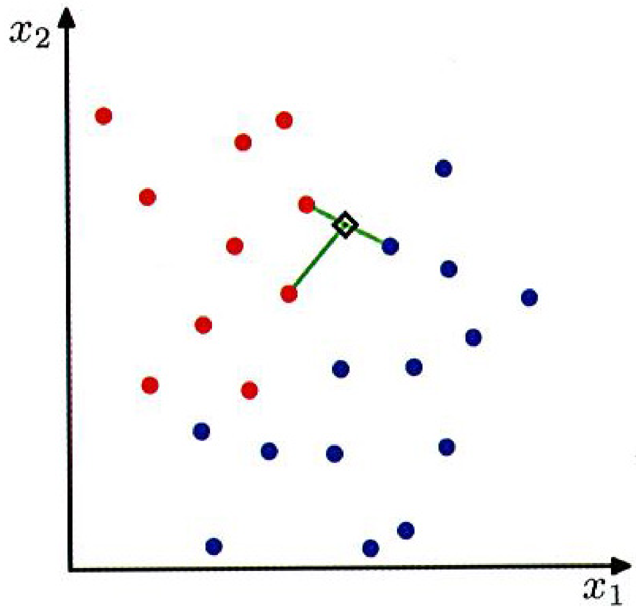
Figure 1.3: Examples of utility matrices

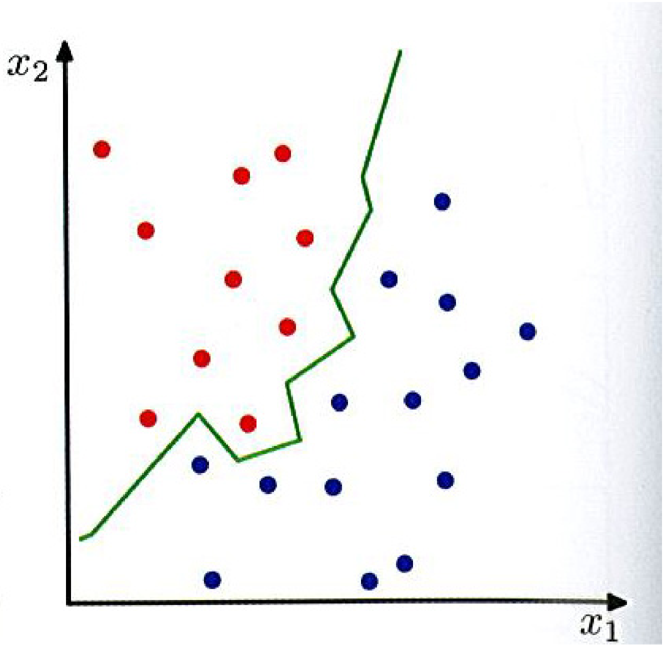
Introduction



(a) Classification (b) Collaborative filtering

Nearest-neighbor methods

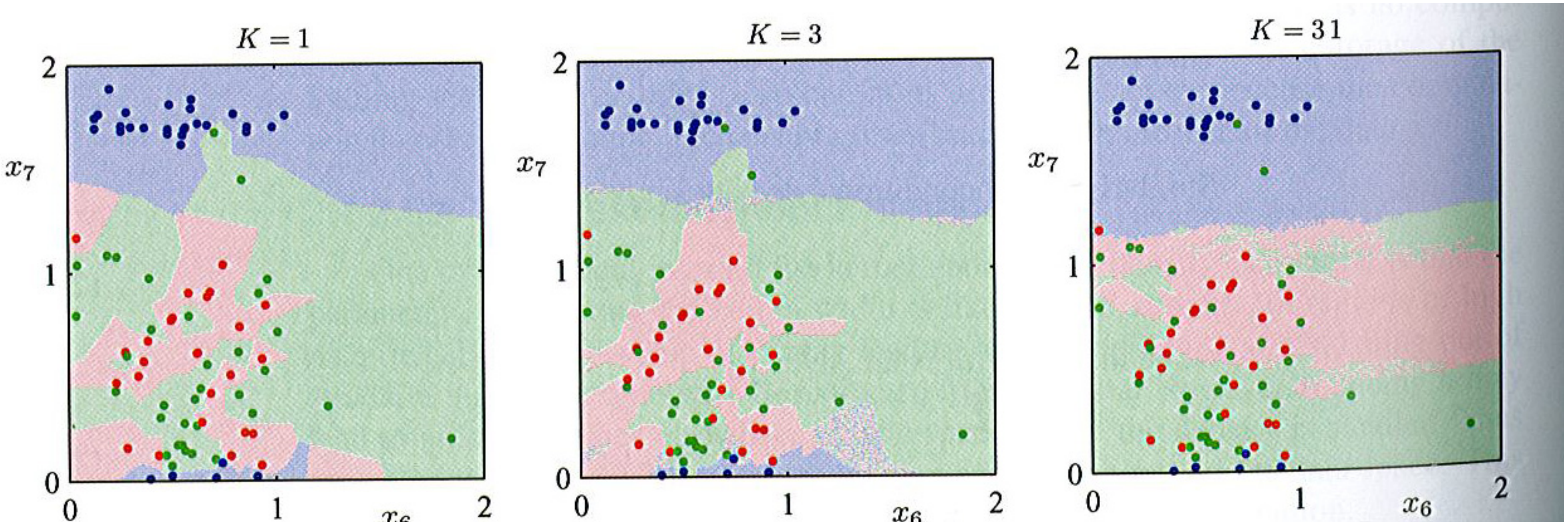




A new point arrives, it is classified according to the majority class membership of its K closest neighbors.

When K=1, the decision boundary is a hyper-plane that form perpendicular bisectors for pairs of points from different classes.

Nearest-neighbor methods



K - pertains to the fit. The k- nearest neighbor fit for $\hat{Y}(x)$ is defined as follows:

$$\hat{Y}(x) = \frac{1}{k} \sum_{x_i \in N_k(x)} y_i.$$

Neighborhood of x defined by the k closes points x_i.

Small K - many small regions.

Larger K - fewer large regions.

Introduction: Collaborative Filtering

Neighborhood-based (memory-based) Collaborative Filtering:

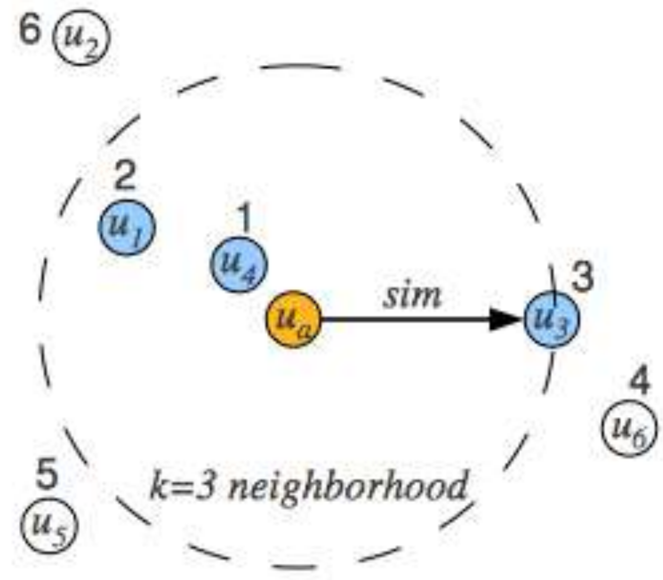
- User-based collaborative filtering:
  - Similar users to “A” are used to make recommendations to “A”.

- Item-based collaborative filtering:
  To make recommendations for a “target item” B,
  - we have to determine the set of items, “S”, most close to “B”.
  - To predict ratings for “B” to a user “A”, we need to aggregate information for those items in “S” across “A”.

Terminology

	i_1	i_2	i_3	i_4	i_5	i_6	i_7	i_8
u_1	?	4.0	4.0	2.0	1.0	2.0	?	?
u_2	3.0	?	?	?	5.0	1.0	?	?
u_3	3.0	?	?	3.0	2.0	2.0	?	3.0
u_4	4.0	?	?	2.0	1.0	1.0	2.0	4.0
u_5	1.0	1.0	?	?	?	?	?	1.0
u_6	?	1.0	?	?	1.0	1.0	?	1.0
u_a	?	?	4.0	3.0	?	1.0	?	5.0
r_a	3.5	4.0			1.3		2.0	

(a)



(b)

ure 1: User-based collaborative filtering example with (a) rating matrix and estimated ngs for the active user, and (b) user neighborhood formation.

Terminology

Long tail – can explain in part the advantages of online retailers compared to traditional “brick and mortar stores”.

Large bookstore at the corner -> thousands of books

Bookstore online -> millions of books

Physical newspaper -> limited articles

Online news -> unlimited articles

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Shaped by aggregate numbers
A finite number of suggestions

Forced to make recommendations
Can’t show everything.

Introduction

Different Flavors Ratings Matrix:

- Continuous ratings: e.g., in a range [1,10]
- Interval ratings: e.g., [1:5] – assumed distance
- Ordinal ratings: e.g., poor, good, very good, excellent
- Binary ratings: e.g., like or dislike.
- Unary ratings: positive affirmation possible, but not negative.

e.g., a "like" in facebook

Introduction

Long Tail:

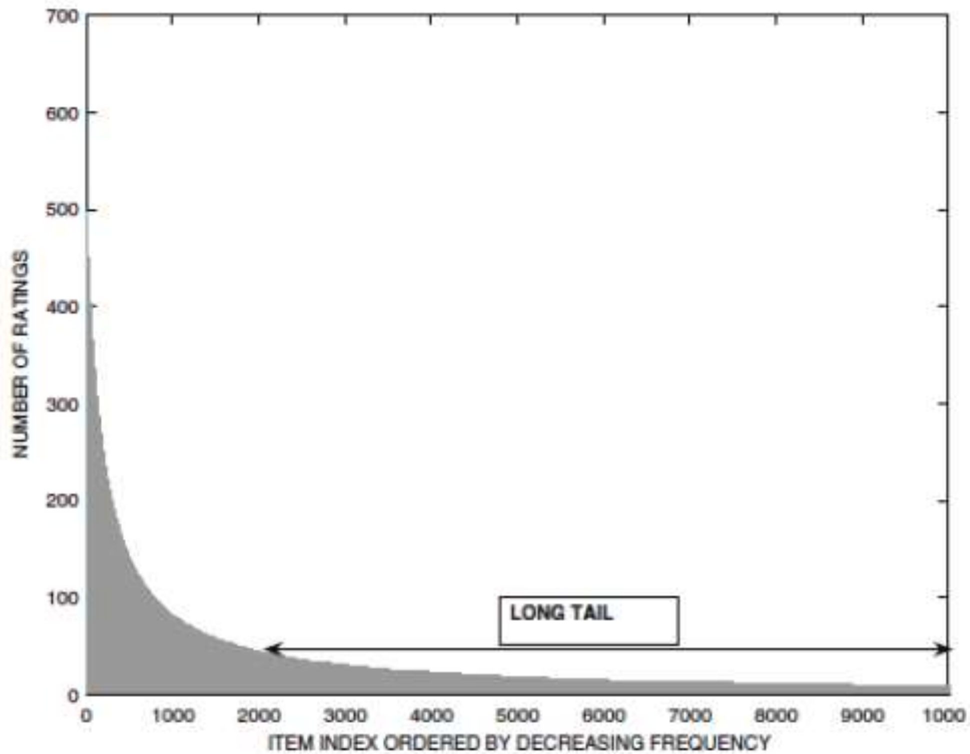


Figure 2.1: The long tail of rating frequencies

Introduction

Problems caused by the "long tail":
Only a small number of items are rated frequently.

Consequently:

1. High-frequency items tend to be competitive, and hold little profit margin. May be beneficial to recommend in the tail.
2. Because of the rarity of items in the tail, the tail items are harder to get robust recommendations for.
3. "Neighborhoods" tend to be formed based on commonalities in the frequently rated items.
4. Overall differences ”popular" and "rare" item ratings are problematic in recommender systems.

Into Thin Air and *Touching the Void*

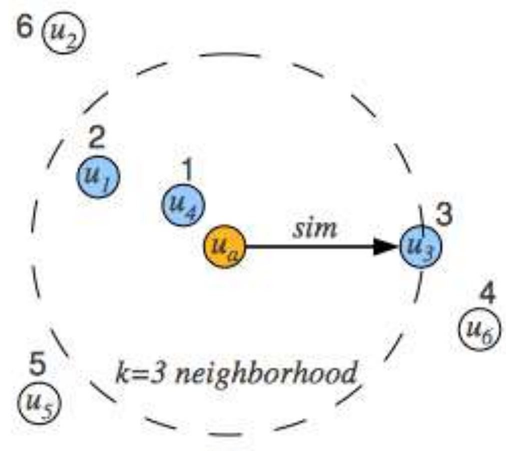
An extreme example of how the long tail, together with a well designed recommendation system can influence events is the story told by Chris Anderson about a book called *Touching the Void*. This mountain-climbing book was not a big seller in its day, but many years after it was published, another book on the same topic, called *Into Thin Air* was published. Amazon's recommendation system noticed a few people who bought both books, and started recommending *Touching the Void* to people who bought, or were considering, *Into Thin Air*. Had there been no on-line bookseller, *Touching the Void* might never have been seen by potential buyers, but in the on-line world, *Touching the Void* eventually became very popular in its own right, in fact, more so than *Into Thin Air*.

User-based CF

Identification of “similar” users.

	i_1	i_2	i_3	i_4	i_5	i_6	i_7	i_8
u_1	?	4.0	4.0	2.0	1.0	2.0	?	?
u_2	3.0	?	?	?	5.0	1.0	?	?
u_3	3.0	?	?	3.0	2.0	2.0	?	3.0
u_4	4.0	?	?	2.0	1.0	1.0	2.0	4.0
u_5	1.0	1.0	?	?	?	?	?	1.0
u_6	?	1.0	?	?	1.0	1.0	?	1.0
u_a	?	?	4.0	3.0	?	1.0	?	5.0
r_a	3.5	4.0			1.3		2.0	

(a)



(b)

ure 1: User-based collaborative filtering example with (a) rating matrix and estimated ngs for the active user, and (b) user neighborhood formation.

May be problematic due to ‘inherent’ differences in users.

User-based CF

Terminology:

- $R = [r_{uj}]$ Be a m x n rating matrix, with “m” users and “n” items
- I_u be the set of item indices for which ratings have been specified by a user. For example, if item 1,3 and 5 are rated by a user: $I_u = \{1, 3, 5\}$

User-based CF

The mean ratings for a user, u:

$$\mu_u = \frac{\sum_{k \in I_u} r_{uk}}{|I_u|} \quad \forall u \in \{1 \ldots m\}$$

The Pearson correlation coefficient between users, u, and v:

$$\text{Sim}(u, v) = \text{Pearson}(u, v) = \frac{\sum_{k \in I_u \cap I_v} (r_{uk} - \mu_u) \cdot (r_{vk} - \mu_v)}{\sqrt{\sum_{k \in I_u \cap I_v} (r_{uk} - \mu_u)^2} \cdot \sqrt{\sum_{k \in I_u \cap I_v} (r_{vk} - \mu_v)^2}}$$

**Note the differences in the mean ratings in these definitions.

User-based CF

Identification of the neighborhood:

$$\mathrm{Sim}(u,v) = \mathrm{Pearson}(u,v) = \frac{\sum_{k \in I_u \cap I_v} (r_{uk} - \mu_u) \cdot (r_{vk} - \mu_v)}{\sqrt{\sum_{k \in I_u \cap I_v} (r_{uk} - \mu_u)^2} \cdot \sqrt{\sum_{k \in I_u \cap I_v} (r_{vk} - \mu_v)^2}}$$

The sets of items/ratings varies so much, that this “top k neighborhood of users” may not be informative for a given item.

Thus, in practice, this may need to be done on a by item basis.

User-based CF

User bias:
dealt with by mean centering.

$$s_{uj} = r_{uj} - \mu_u \quad \forall u \in \{1 \ldots m\}$$

When the final rating is given, it has to be "adjusted":

$$\hat{r}_{uj} = \mu_u + \frac{\sum_{v \in P_u(j)} \text{Sim}(u,v) \cdot s_{vj}}{\sum_{v \in P_u(j)} |\text{Sim}(u,v)|} = \mu_u + \frac{\sum_{v \in P_u(j)} \text{Sim}(u,v) \cdot (r_{vj} - \mu_v)}{\sum_{v \in P_u(j)} |\text{Sim}(u,v)|}$$

User-based CF

Example:

5 users

6 items

Ratings range: {1,2,3,4,5,6,7}

User-based CF

Example: Target user 3

5 users

6 items

Ratings range: {1,2,3,4,5,6,7}

Table 2.1: User-user similarity computation between user 3 and other users

Item-Id ⇒ / User-Id ⇓	1	2	3	4	5	6	Mean Rating	Cosine(i, 3) (user-user)	Pearson(i, 3) (user-user)
1	7	6	7	4	5	4	5.5	0.956	0.894
2	6	7	?	4	3	4	4.8	0.981	0.939
3	?	3	3	1	1	?	2	1.0	1.0
4	1	2	2	3	3	4	2.5	0.789	-1.0
5	1	?	1	2	3	3	2	0.645	-0.817

User-based CF

$$\text{Cosine}(1,3) = \frac{6*3+7*3+4*1+5*1}{\sqrt{6^2+7^2+4^2+5^2}\cdot\sqrt{3^2+3^2+1^2+1^2}} = 0.956$$

$$\text{Pearson}(1,3) =$$

$$= \frac{(6-5.5)*(3-2)+(7-5.5)*(3-2)+(4-5.5)*(1-2)+(5-5.5)*(1-2)}{\sqrt{1.5^2+1.5^2+(-1.5)^2+(-0.5)^2}\cdot\sqrt{1^2+1^2+(-1)^2+(-1)^2}}$$

$$= 0.894$$

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Item-Id ⇒ / User-Id ⇓	1	2	3	4	5	6	Mean Rating	Cosine(i, 3) (user-user)	Pearson(i, 3) (user-user)
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2	6	7	?	4	3	4	4.8	0.981	0.939
3	?	3	3	1	1	?	2	1.0	1.0
4	1	2	2	3	3	4	2.5	0.789	-1.0
5	1	?	1	2	3	3	2	0.645	-0.817

User-based CF

By Pearson, raw:

$$\hat{r}_{31} = \frac{7 * 0.894 + 6 * 0.939}{0.894 + 0.939} \approx 6.49$$

$$\hat{r}_{36} = \frac{4 * 0.894 + 4 * 0.939}{0.894 + 0.939} = 4$$

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Item-Id ⇒ / User-Id ⇓	1	2	3	4	5	6	Mean Rating	Cosine(i, 3) (user-user)	Pearson(i, 3) (user-user)
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3	?	3	3	1	1	?	2	1.0	1.0
4	1	2	2	3	3	4	2.5	0.789	-1.0
5	1	?	1	2	3	3	2	0.645	-0.817

User-based CF

By Pearson, raw:

$$\hat{r}_{31} = 2 + \frac{1.5 * 0.894 + 1.2 * 0.939}{0.894 + 0.939} \approx 3.35$$

$$\hat{r}_{36} = 2 + \frac{-1.5 * 0.894 - 0.8 * 0.939}{0.894 + 0.939} \approx 0.86$$

Table 2.1: User-user similarity computation between user 3 and other users

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2	6	7	?	4	3	4	4.8	0.981	0.939
3	?	3	3	1	1	?	2	1.0	1.0
4	1	2	2	3	3	4	2.5	0.789	-1.0
5	1	?	1	2	3	3	2	0.645	-0.817

User-based CF

Alternative similarity functions (select):

$$\mathrm{DiscountedSim}(u,v) = \mathrm{Sim}(u,v) \cdot \frac{\min\{|I_u \cap I_v|, \beta\}}{\boxed{\beta}}$$

Threshold paramater

User-based CF

Alternative for prediction function:

$$\hat{r}_{uj} = \mu_u + \sigma_u \frac{\sum_{v \in P_u(j)} \text{Sim}(u, v) \cdot z_{vj}}{\sum_{v \in P_u(j)} |\text{Sim}(u, v)|}$$

where,

$$\sigma_u = \sqrt{\frac{\sum_{j \in I_u} (r_{uj} - \mu_u)^2}{|I_u| - 1}} \quad \forall u \in \qquad z_{uj} = \frac{r_{uj} - \mu_u}{\sigma_u} = \frac{s_{uj}}{\sigma_u}$$

**conflicting results on the benefit form z-score.

User-based CF

Amplification/ dampening:

$$\hat{r}_{uj} = \mu_u + \frac{\sum_{v \in P_u(j)} \boxed{\mathrm{Sim}(u,v)} \cdot s_{vj}}{\sum_{v \in P_u(j)} |\boxed{\mathrm{Sim}(u,v)}|} = \mu_u + \frac{\sum_{v \in P_u(j)} \boxed{\mathrm{Sim}(u,v)} \cdot (r_{vj} - \mu_v)}{\sum_{v \in P_u(j)} |\boxed{\mathrm{Sim}(u,v)}|}$$

where,

$$\mathrm{Sim}(u,v) = \mathrm{Pearson}(u,v)^{\alpha}$$

for

$$\alpha > 1,$$

User-based CF

Long tail revisited:

- Popular items common across users, may worsen the predictive value, not discrimitive.
- Similar ideas in information retrieval.
- Inverse ratings for and object, j,:

$$w_j = \log\left(\frac{m}{m_j}\right) \quad \forall j \in \{1 \ldots n\}$$

$$\text{Pearson}(u, v) = \frac{\sum_{k \in I_u \cap I_v} w_k \cdot (r_{uk} - \mu_u) \cdot (r_{vk} - \mu_v)}{\sqrt{\sum_{k \in I_u \cap I_v} w_k \cdot (r_{uk} - \mu_u)^2} \cdot \sqrt{\sum_{k \in I_u \cap I_v} w_k \cdot (r_{vk} - \mu_v)^2}}$$

Item-based CF

- Similarities computed between columns (items).
- User-bias corrected – mean center rows.

Let U_i be the set of user that have rated item "i".

E.g., Suppose users 1,3 and 4 have rated the item:

$$U_i = \{1, 3, 4\}.$$

Item-based CF

- The adjusted cosine similarity between items "i" and "j", is defined as:

$$\text{AdjustedCosine}(i, j) = \frac{\sum_{u \in U_i \cap U_j} s_{ui} \cdot s_{uj}}{\sqrt{\sum_{u \in U_i \cap U_j} s_{ui}^2} \cdot \sqrt{\sum_{u \in U_i \cap U_j} s_{uj}^2}}$$

**Pearson can be user but does not work as well in practice.

Item-based CF

Suppose you need the rating of target item "t" for a user "u".

1. Determine the "top-k" most similar items to item t, based on the adjusted cosine.
2. Identify the items on this list that the user has rated, $Q_t(u)$.
3. The weighted value of these items is the prediction.

$$\hat{r}_{ut} = \frac{\sum_{j \in Q_t(u)} \text{AdjustedCosine}(j,t) \cdot r_{uj}}{\sum_{j \in Q_t(u)} |\text{AdjustedCosine}(j,t)|}$$

* Use user's own ratings on similar items!!

Item-based CF

Example:

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3	?	3	3	1	1	?	2	1.0	1.0
4	1	2	2	3	3	4	2.5	0.789	-1.0
5	1	?	1	2	3	3	2	0.645	-0.817

Item-based CF

Example:

Table 2.2: Ratings matrix of Table 2.1 with mean-centering for adjusted cosine similarity computation among items. The adjusted cosine similarities of items 1 and 6 with other items are shown in the last two rows.

Item-Id ⇒ / User-Id ⇓	1	2	3	4	5	6
1	1.5	0.5	1.5	-1.5	-0.5	-1.5
2	1.2	2.2	?	-0.8	-1.8	-0.8
3	?	1	1	-1	-1	?
4	-1.5	-0.5	-0.5	0.5	0.5	1.5
5	-1	?	-1	0	1	1
Cosine$(1, j)$ (item-item)	1	0.735	0.912	-0.848	-0.813	-0.990
Cosine$(6, j)$ (item-item)	-0.990	-0.622	-0.912	0.829	0.730	1

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2	1.2	2.2	?	-0.8	-1.8	-0.8
3	?	1	1	-1	-1	?
4	-1.5	-0.5	-0.5	0.5	0.5	1.5
5	-1	?	-1	0	1	1
Cosine(1, j) (item-item)	1	0.735	0.912	-0.848	-0.813	-0.990
Cosine(6, j) (item-item)	-0.990	-0.622	-0.912	0.829	0.730	1

$$\text{AdjustedCosine}(1,3) = \frac{1.5 * 1.5 + (-1.5) * (-0.5) + (-1) * (-1)}{\sqrt{1.5^2 + (-1.5)^2 + (-1)^2} \cdot \sqrt{1.5^2 + (-0.5)^2 + (-1)^2}} = 0.912$$

$$\hat{r}_{31} = \frac{3 * 0.735 + 3 * 0.912}{0.735 + 0.912} = 3$$

$$\hat{r}_{36} = \frac{1 * 0.829 + 1 * 0.730}{0.829 + 0.730} = 1$$

Collaborative Filtering 0-1

Less information available on 0-1:

- People don’t want to do ratings.
- Preference has to be inferred by analyzing usage behavior.

In this setting, the 1’s are easy to understand!

Collaborative Filtering 0-1

Less information available on 0-1:

- People don't want to do ratings.
- Preference has to be inferred by analyzing usage behavior.

In this setting, the 1's are easy to understand!

The zeros are more mysterious:

S1) The customer did not need the product right now.

S2) The customer did not know about the product.

S3) The customer did not like the product.

Same issues hold in "click-stream" modeling.

Collaborative Filtering 0-1

In the 0-1 case with $r_{jk} \in 0, 1$ where we define:

$$r_{jk} = \begin{cases} 1 & \text{user } u_j \text{ is known to have a preference for item } i_k \\ 0 & \text{otherwise.} \end{cases}$$

Two possibilities:

- Assume “missing”.
- Assume “negative”.
- Some trade-off’s have been proposed.

$$\mathrm{sim}_{\mathrm{Jaccard}}(\mathcal{X}, \mathcal{Y}) = \frac{|\mathcal{X} \cap \mathcal{Y}|}{|\mathcal{X} \cup \mathcal{Y}|}, \tag{7}$$

where $\mathcal{X}$ and $\mathcal{Y}$ are the sets of the items with a 1 in user profiles u_a and u_b, respectively.